

# INF1: HS1-Radioastronomie

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## Abstract

In the experiment **HS1-Radioastronomy** the rotational curve of the Milky Way is determined and thus to calculate the mass, as well as the composition of the Milky Way. Furthermore, the background radiation and spatial resolving power of the telescope are determined. This is done with the help of a radio telescope, which is located at the Sternenswarte Bergedorf, Hamburg.

# 1 Introduction

The goal of the experiment is to find neutral hydrogen in the Milky Way by measuring it as accurately as possible and to calculate its Doppler shift. With this, the speed of the movement, as well as its direction, can then be concluded. With this data, the mass in the Milky Way can be determined in relation to the distance to the galaxy center.

The HS1 radio astronomy experiment is divided into four main sections and six subsections:

- Calibration
- Determination of the spatial resolving power
- Determination of the sensitivity of the instrument
- Observation of neutral hydrogen in the Milky Way
  - Determination of the spectra, as well as discussion about deviations
  - Comparison of spectra with the HI line profile of the Leiden-Argentine-Bonn (LAB) survey, and discussion of discrepancies.
  - Calculation of the max. radial velocity, as well as the estimation of the inaccuracy.
  - Calculation of the path velocities at the respective tangential points
  - Generation of a graphical rotation curve
  - Calculation of the mass of the Milky Way

With the first 3 points calibration, determination of the spatial resolving power and determination of the sensitivity of the instrument, the aim is to determine and to calculate out possible disturbances, so that the measurement of the neutral hydrogen can be accomplished as accurately as possible. The fourth point is the actual measurement, as well as all calculations to determine the radial and orbital velocity, as well as the mass of the Milky Way.

## 1.0.1 Radio waves

The range of the radio radiation  $\lambda=0.5\text{ mm} - 20\text{ mm}$  is particularly well suited to measure the Milky Way, since the radiation cannot be blocked by dust clouds. This allows deeper insights into the Milky Way, which would not be possible in the optical range.

## 1.1 21cm line of neutral hydrogen atom

Hydrogen is the primordial element of the universe and was formed shortly after the Big Bang. It is the element that is most abundant in the universe and thus is also found a lot in the Milky Way.

The state form of hydrogen is temperature dependent, being in atomic form (H1) from 10 K to 10000 K, ionized at higher temperatures being in molecular form at temperatures  $< 10$  K.

The characteristic 21 cm lines of hydrogen result from the hyperfine structure transition, which is at a rest frequency of  $\nu_0 = 1420.406$  MHz in the ground state. The electron can assume either a parallel or anti-parallel magnetic moment relative to the proton. The state of the anti-parallel is more energetic and the 21cm lines are generated.(see figure [1])

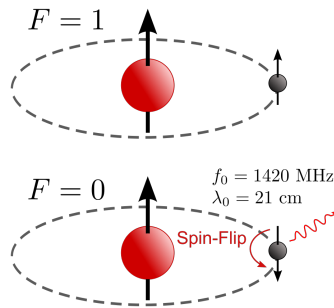


Figure 1: Hydrogen Spin Flip and the 21cm emission. Figure taken from [1].

This transition occurs only every  $1.1 \cdot 10^7$  years. However, this time decreases to 300-400 years due to collisions in the interstellar space. But as already mentioned, hydrogen is the most frequent element in the Milky Way, so nevertheless the 21cm lines can be measured relatively often.

### 1.1.1 Emitted electromagnetic waves and Doppler effect

Due to the relatively simple structure of the atom, which consists of only one proton and one electron, a transition from the parallel to the lower-energy anti-parallel spin state occurs, emitting characteristic wavelengths of 21 cm, which can be received from the entire Milky Way region.

Thereby the 21 cm lines are shifted in frequency by the Doppler effect. The received light is shifted to red when the hydrogen moves away from the observer and to blue when the hydrogen moves towards the observer. The Doppler shift occurs because the light gains or loses additional energy when emitted by a moving object. As the photon moves in the direction of the motion of the object that emitted the photon, kinetic energy is added to the photon. Since the speed of light is constant in all reference frames, the energy cannot be translated into additional speed. Instead, the total energy of the photon is increased

and thus also its frequency. If the photon moves against the direction of the object that emitted the photon, it loses energy and thus also decreases the frequency. This is called relativistic Doppler effect, graphically represented in figure 2.

$$f_B = f_s \frac{\sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{v}{c}} \quad (1.1.1)$$

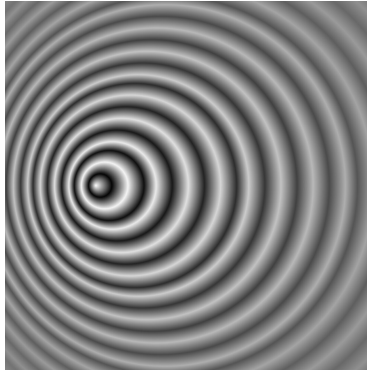


Figure 2: Representation of the Doppler effect [2]

## 2 Observation, evaluation and results

As the different parts of the experiment were executed on different days, each part includes the observation and evaluation as well as the result to achieve a better overview of the experiment.

For our measurements the radio telescope SPIDER 300A is used. This is mounted on the roof of the observatory in Bergedorf. This telescope operates in the frequency range from 15 MHz to 600 GHz. The telescope is a parabolic antenna, which stands in azimuthal alignment to the sky and has a diameter of 3 meters. It is possible to adjust the telescope in horizontal and vertical direction for optimal observation of the sky. The alignment is done by a computer and does not require any manual adjustment as it used to be the case with optical telescopes. The SPIDER 300A is limited in its spatial extent, since this is proportional to the wavelength ( $\lambda \sim 1$  m). The diffraction limit of the telescope, describes the smallest angular distance between 2 objects, which can be detected. The spatial resolution is assumed to be the half-width  $q$  of the central diffraction disk of the telescope:

$$\Omega_A \propto q^2 \approx \frac{\lambda^2}{D^2} \quad (2.0.1)$$

$$q = 1.02 \frac{\lambda}{D} \text{rad} = 58.4^\circ \frac{\lambda}{D} \quad (2.0.2)$$

## 2.1 Calibration of the telescope (28.03.2022)

This series of measurements was carried out on 28.03.2022.

### 2.1.1 Weather observation on date: 28.03.2022

The sky was cloudy, but without rain. It was about 10° C and the wind was moderate to calm. The visibility was good.

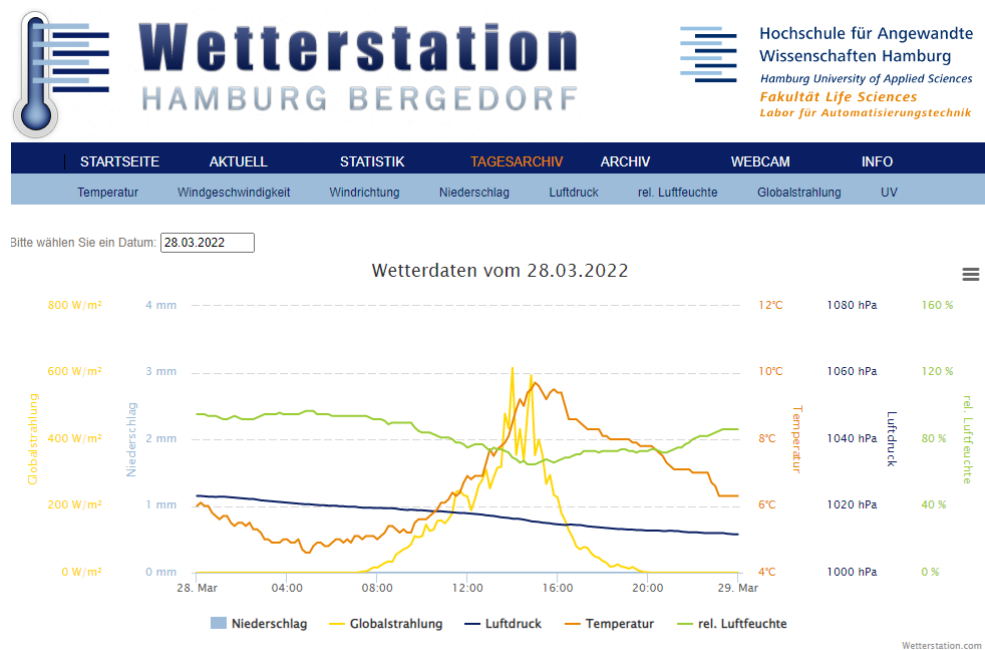


Figure 3: weather observation on 28.03.22

## 2.2 Pointing correction (28.03.2022)

To perform an alignment correction, the position of the sun must be determined precisely. For this purpose, the antenna automatically adjusts to a calculated position, which depends on the time of day, and records the power (TP value) at the receiver. The highest power (approx. 20000) at which the sun must be located is then searched for, assuming it to be a point source. To do this, the longitude and latitude angles are manually adjusted until the maximum is found.

The time of measurement was 15:50. The preset Position of the sun at the time of measurement was at Azimuth: 169° 13' Elevation: 39° 07'.

The maxima were found in the following positions:

|            | longitude | latitude |
|------------|-----------|----------|
| 1. setting | 2         | 0        |
| 2. setting | 4         | -1       |
| 3. setting | 3         | 0        |

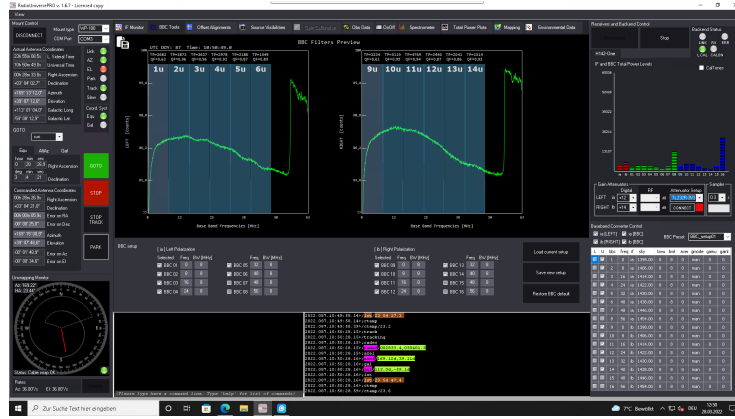


Figure 4: Illustration of a screenshot of the program in the BBC Tools area, with which the correction was made on the basis of the sun. The TP value determines the power that must be maximized. It can be seen that the bands 7u and 8u, as well as 15u and 16u were not used, because they are a source of interference.



## 2.3 Weather observation on date: 21.04.2022

The following section of the experiment was carried out on another day. The sky was cloudy, but without rain. It was about 12° C and the wind was moderate. The visibility was okay.

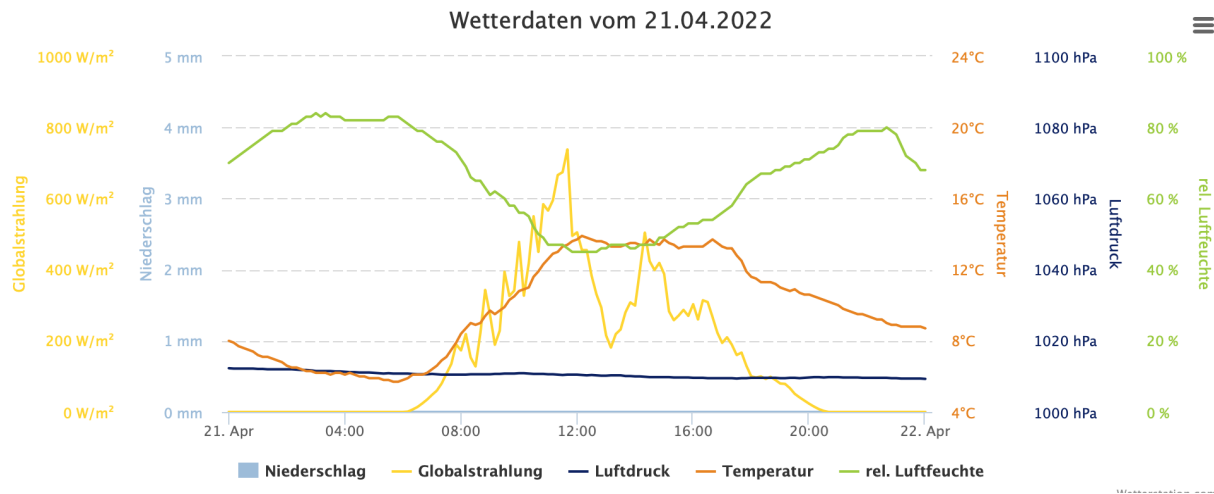


Figure 5: weather observation on 21.04.22. Data taken from [3]

The pointing correction is already explained in the previous section.

The measurement log for this day can be found in the appendix as figure 15.

### 2.3.1 Determination of the system temperature

The receiving radiation, consists not only of the radiation from the source under investigation, but is also composed of ambient radiation and the antenna's own radiation. These sources of interference must be calculated out in order to obtain an optimal measurement. The antenna temperature  $T_A$  is composed of the following components:

$$T_A = T_s + T_h + T_{\text{rec}} = T_s + T_{\text{sys}} \quad (2.3.1)$$

Now an ON-OFF series of measurements is performed to determine the interfering factors. With a stable telescope, the sun is measured first and then the radiation of the background. The theoretical value of the beam width is 4 °.

Now the system temperature on the left  $T_{\text{sys,left}}$  and the system temperature on the right  $T_{\text{sys,right}}$ , as well as the solar temperature  $T_{\text{sun}}$  are calculated.

$$T_{\text{sys}} = T_{\text{cal}} \frac{P_{\text{off}}}{P_{\text{cal}} - P_{\text{off}}} \quad (2.3.2)$$

The calibration temperature is  $T_{\text{cal}}=170$  K.

$$T_{\text{sun}} = T_{\text{sys}} \frac{P_{\text{on}} - P_{\text{off}}}{P_{\text{off}}} \quad (2.3.3)$$

| Number of BBC  | On_tpi   | Off_tpi            | Cal_tpi            | T_sys,left in K    | T_sys,right in K | T_Sun in K  |
|----------------|----------|--------------------|--------------------|--------------------|------------------|-------------|
| 1              | 23141,66 | 1252,5             | 3776,64            | 84,35546364        |                  | 1474,227737 |
| 2              | 31544,24 | 1829,46            | 5518,92            | 84,2964011         |                  | 1369,173971 |
| 3              | 30563,1  | 1781,78            | 5323,36            | 85,52753291        |                  | 1381,537167 |
| 4              | 26926,14 | 1511,28            | 4261,48            | 93,4177878         |                  | 1570,986183 |
| 5              | 21204,62 | 1171,18            | 2994,1             | 109,2207009        |                  | 1868,25796  |
| 6              | 15997,4  | 899,1              | 2216,38            | 116,0322786        |                  | 1948,493107 |
| 9              | 25041,36 | 1245,66            | 4129,76            |                    | 73,42401442      | 1402,61052  |
| 10             | 38401,6  | 1978,12            | 6727,86            |                    | 70,79974904      | 1303,648537 |
| 11             | 37315,4  | 1942,94            | 6517,98            |                    | 72,19604637      | 1314,375    |
| 12             | 28058,06 | 1446,68            | 4540,22            |                    | 79,4997317       | 1462,381156 |
| 13             | 17339,48 | 934,84             | 2604,68            |                    | 95,17247161      | 1670,093422 |
| 14             | 13272,44 | 762,66             | 1909,14            |                    | 113,0871886      | 1854,949585 |
|                |          |                    |                    |                    |                  |             |
| <b>Average</b> |          | <b>95,47502749</b> | <b>84,02986696</b> | <b>1551,727862</b> |                  |             |
| <b>Error</b>   |          | <b>13,88045664</b> | <b>16,83191696</b> | <b>229,6903603</b> |                  |             |

Figure 6: Table for TP values of the on-off measurement for the left and right polarized BBCs, as well as TP values for the sun.

The temperatures are determined as the following:

- $T_{\text{sys,left}} = 95.5 \pm 13.9$  K
- $T_{\text{sys,right}} = 84.0 \pm 16.8$  K
- $T_{\text{sun}} = 1551.7 \pm 229.7$  K

To measure the antenna temperature of the sun as a function of the flux density, the following formula is used:

$$S_v = \frac{T_A}{A_e} \cdot 2k_B \quad (2.3.4)$$

The effective antenna area  $A_e$  is calculated with the following formula:

$$A_e = \eta A_g = 0.5\pi R^2 \quad (2.3.5)$$

and with  $\eta = 0.5$  and  $R = 1,5$  m.

From this follows:

$$S_v = \frac{T_A \cdot 2k_B}{0.5\pi R^2} \quad (2.3.6)$$

Now the temperatures are converted to the flux density, as shown for the sun:

$$9,3411 \cdot 10^{-21} \frac{W}{m \cdot Hz} \cdot 10^{26} = 934112.1 \text{ Jy} \quad (2.3.7)$$

This is done for every temperature.

The flux density is determined as the following:

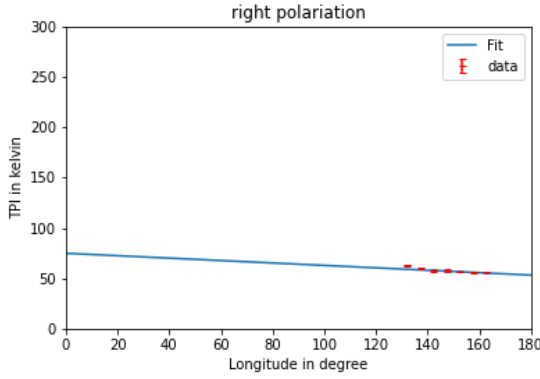
- $S_{v_{\text{sys, left}}} = 74558.3 \pm 8339.9 \text{ Jy}$
- $S_{v_{\text{sys, right}}} = 65620.6 \pm 13144.4 \text{ Jy}$
- $S_{v_{\text{sun}}} = 1211775.3 \pm 179369.8 \text{ Jy}$

The left and right system temperatures are within the limits of the error. The calibration is therefore verified.

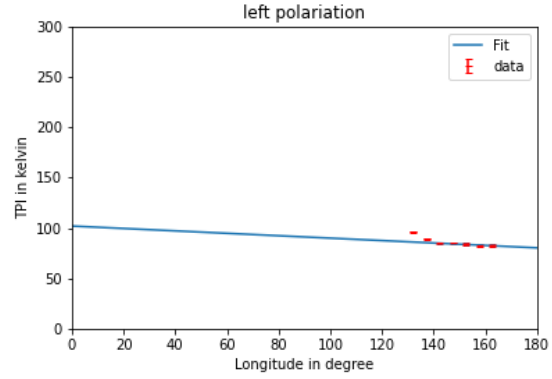
This can now be compared to the data from the World Data Center [4], which has a theoretical value of  $T_{\text{sun}}^{\text{th}} = 1.3 \cdot 10^6 \text{ Jy}$ . This is within limits of the error.

## 2.4 Background radiation (21.04.2022)

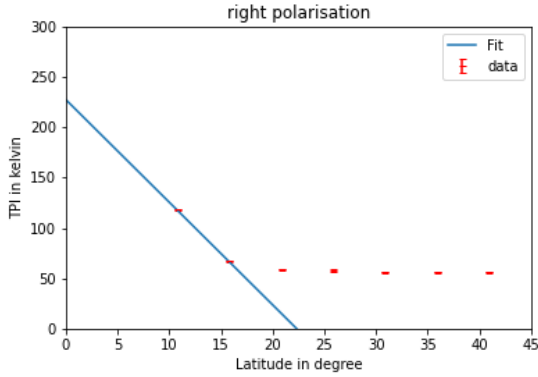
As there is no radio source in the field of vision, the antenna temperature  $T_A$  is assumed to be equal to the system temperature  $T_{\text{sys}}$ . Therefore, the antenna temperature is measured for different azimuth values going in steps of five degrees. After that, the same is done for different values of elevation. This is done for both right and left circular polarisation. Four graphs (see figure 7) can be determined. For that, they are normalized to the system temperature. They can also be extrapolated up to an elevation of  $0^\circ$ .



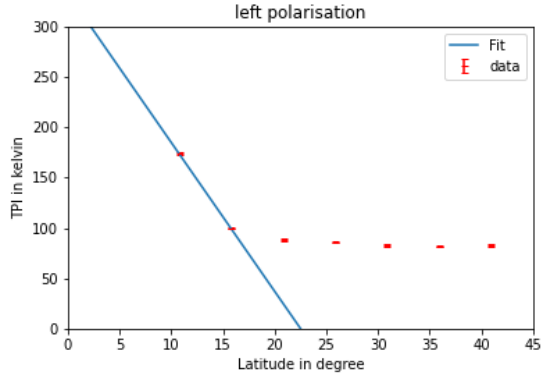
(a) Longitude (azimuth) right circular polarized (other scaling: figure 18)



(b) Longitude (azimuth) left circular polarized (other scaling: figure 18)



(c) Latitude (elevation) right circular polarized



(d) Latitude (elevation) left circular polarized

Figure 7: Cross scan over Orion A, showing the temperature in Kelvin on the y-axis and the offset from orion in degree on the x-axis. From that, the temperature at 0 K can be determined. The mean for the longitude is  $88.43 \pm 3.46$  K, while for the latitude it is  $279.59 \pm x$  K.

A linear fit is done for either the first two data points of the latitude or the last five data points of the longitude. The results of the linear fits are in table 1.

| Measurement            | y-intercept in K  | slope in K/degree | $\chi^2$ -values |
|------------------------|-------------------|-------------------|------------------|
| <b>Latitude left</b>   | $332.17 \pm x$    | $-14.74 \pm x$    | 53143.6          |
| <b>Latitude right</b>  | $227.01 \pm x$    | $-10.16 \pm x$    | 25489.4          |
| <b>Longitude left</b>  | $101.93 \pm 3.46$ | $-0.12 \pm 0.02$  | 20.56            |
| <b>Longitude right</b> | $74.93 \pm 3.46$  | $-0.12 \pm 0.02$  | 2.96             |

Table 1: Table including the FWHM and  $\chi^2$ -values of the cross scanning of the sun for the longitude, latitude and both circularisations. The average is also calculated. As there are only two data points used for the estimation of the linear fit of the latitude, the error  $x$  is very high, but can not be estimated correctly. It has to be seen as a rough estimation.

For the longitude, the estimated temperature at 0 K for the left circularisation is  $101.93 \pm 3.46$ , for the right circularisation it is  $74.93 \pm 3.46$ . The  $\chi^2$ -values are 20.56 and 2.96 respectively. This estimation is very far away from the theoretical value. This is mostly, because the evaluation method is not very precise in the first place. By choosing the number of data points, that have to be interpolated, one can clearly change the value of the y-intercept in the linear regression. As the data shows the offset from orion on the x-axis, the temperature is higher, when it is closer to orion. However, the radio radiation of the earth, does not really have an effect for the azimuthal measurements, as they are not interfered by the earths radiation as much, as they are above the  $15^\circ$ .

For the latitude the temperature at different elevations is measured. The values of the temperature are influenced by the atmosphere of the earth, as this has a temperature. So the closer the telescope is directed towards the ground, the closer it is to that temperature. They are relatively constant up to  $15^\circ$ . For degrees closer to the earth, the temperature gets closer to the black-body-radiation temperature of the earth of  $T_{\text{earth}} = 252$  K [5]. Since the telescope cannot go under an elevation of  $10^\circ$ , the influence of the earth's radio radiation at low elevations has to be extrapolated. The first two points are used, because they seem to increase towards lower latitudes.

The temperature estimated for the left circularisation is  $332.17 \pm x$ , for the right circularisation it is  $227.01 \pm x$ . The error is high, but can not be estimated, as it is only a fit between two data points and therefore, mathematically, there is no error. These estimated temperatures are only a rough estimation as it is only a fit between two data point. This is also shown by the high  $\chi^2$ -values. The average temperature of the two linear fits for both circularisations is 279.59 K, which is closer to the theoretical value than what is expected of a fit of two data points. When it is directed towards the sky however, the measured temperature consists of the background temperature, which is the noise in this experiment.

## 2.5 Determination of the spatial resolving power (21.04.2022)

To determine the spatial resolution, the telescope is directed towards the sun and the antenna temperature  $T_A$  is measured for different azimuth values going in steps of one degree. After that, the same is done for different values of elevation. This is done for both right and left circular polarisation. After converting it to a .text file, four graphs (see figure 8) can be determined. For that, they are normalized to the system temperature.

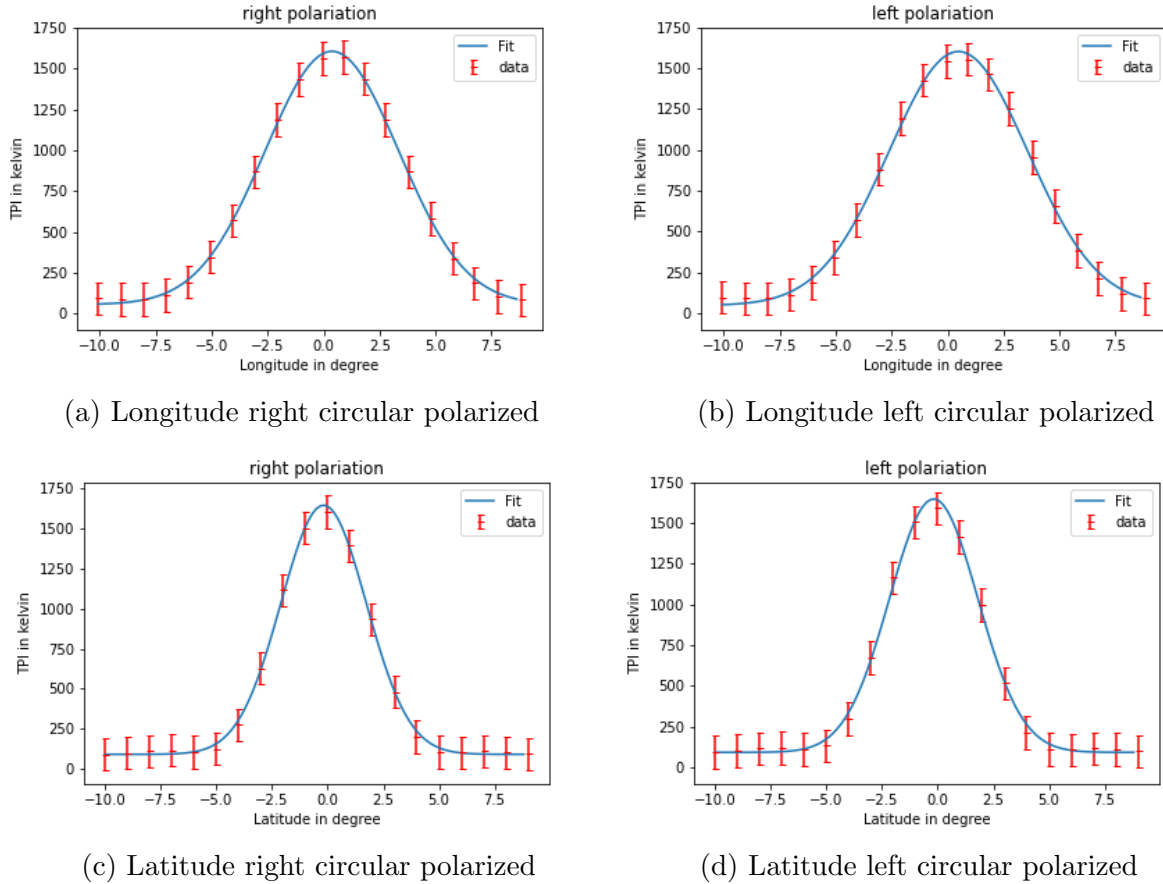


Figure 8: Cross scan over the sun, showing the temperature in Kelvin on the y-axis and the offset from the sun in degree on the x-axis. From that, the FWHM for each can be determined. The average is  $5.34 \pm 0.08$ .

The theoretically expected FWHM is  $q = 1.02\lambda/D$  in radians. As the wavelength  $\lambda = 0.21$  m and the antenna diameter  $D = 3.0$  m, the theoretical value of the FWHM is  $q^{\text{th}} = 0.0714$ . This is equal to  $4.0909^\circ$ .

To determine the FWHM in degree, the sigma value of the gauss fit has to be multiplied by 2.3548. For the data of the longitude another factor has to be considered: As the angle does not describe a great circle in the sky, but more of a latitude in the sky, the angle can

be corrected by approximately  $\cos(32)$ , as the elevation at the time was 32 degrees. The following four spatial resolutions can therefore be determined. The  $\chi^2$ -value is also listed in table 2:

| Measurement            | FWHM in degree | Error in degree | $\chi^2$ -values |
|------------------------|----------------|-----------------|------------------|
| <b>Latitude left</b>   | 4.68           | 0.09            | 0.08             |
| <b>Latitude right</b>  | 4.52           | 0.06            | 0.06             |
| <b>Longitude left</b>  | 6.19           | 0.11            | 0.06             |
| <b>Longitude right</b> | 5.96           | 0.07            | 0.03             |
| <b>Average</b>         | 5.34           | 0.08            | 0.06             |

Table 2: Table including the FWHM and  $\chi^2$ -values of the cross scanning of the sun for the longitude, latitude and both circularisations. The average is also calculated.

The average of  $(5.34 \pm 0.08)^\circ$  is close to the theoretical value of  $4.0909^\circ$ . However, it is not within the error tolerance. The values of the latitude are closer to the actual resolution. This could be because the elevation of the telescope is not measured precise enough. The right polarisation is also slightly closer to the theoretical value. Possible reasons also include the error in the calculation of the system temperature, as well as a difference in temperature of the sun between the two measurements. Further errors are discussed in the overall error discussion.



## 2.6 Determination of the sensitivity of the instrument (21.04.2022)

### 2.6.1 Theoretical sensitivity

The lowest verifiable flux density  $\Delta S_v$  for a given temperature  $T_{\text{sys}}$  for an integration time of 1 s can first be calculated theoretically with the radiometer equation:

$$\Delta S_v = \frac{2k}{A_e} \frac{C_s T_{\text{sys}}}{\sqrt{\Delta \nu \tau}} \quad (2.6.1)$$

$$\Delta S_v = \frac{2k}{0.5 \cdot \pi \cdot 1.5^2 m^2} \frac{2 \cdot 89.8 K}{\sqrt{61000 Hz \cdot 1 s}} \quad (2.6.2)$$

$$\Delta S_v = 2.839 \cdot 10^{-24} \frac{W}{m^2 Hz} = 283.93 \text{Jy} \quad (2.6.3)$$

Then Tau A was chosen as an appropriate source. The necessary integration time  $\tau$  was chosen to detect the selected source with a signal to noise ratio of  $\frac{S}{N} = 10$  in each channel. The flux density of TauA is  $940 \pm 50$  for a wavelength of 0.21 cm for Tau A [6].

The signal to noise ratio is calculated with equation 2.6.4 to be  $3.31 \pm 0.18$  for 1s. To get a signal to noise ratio of over 10, the integration time needs to be  $9.13 \pm 1.00$  s as it is proportional to  $\sqrt{\tau}$ .

$$\frac{S}{N} = \frac{S_v}{\Delta S_v} = \frac{940 \pm 50 \text{Jy}}{283.93 \text{Jy}} = 3.31 \pm 0.18 \quad (2.6.4)$$

### 2.6.2 Uncertainty of radiant flux measurements

The flux density measurement is performed as a relative measurement, comparing a measurement with the source to one without. A beam-switching-measurement is performed and the antenna temperature is measured. The difference gives the antenna temperature of the source  $T_S$ :

$$T_A(ON) - T_A(OFF) = (T_S + T_{sys}) - T_{sys} \quad (2.6.5)$$

First, the approximate flux density  $S_v$  of a mobile phone on the moon recieved by the earth is calculated to then better evaluate the calculated temperatures: The phone emits a power of 2W and has a normal distribution at a frequency of 900 MHz with a FWHM=80kHz ( $\sigma = v = 33.3kHz$ ). The distance D is the distance to the moon:

$$S_v = \frac{P}{4\pi D^2 \cdot \sqrt{2\pi} \cdot \sigma} = \frac{2W}{4\pi(384.4 \cdot 10^6m)^2 \cdot \sqrt{2\pi} \cdot 33.3 \cdot 10^3Hz} \propto 1290Jy \quad (2.6.6)$$

This can then be compared to the flux density of Cas A at 900 MHz. The flux density at 1420.4 MHz is  $1420 \pm 70$ . We assume a spectral progression of  $S_v \propto \nu^\alpha$  with  $\alpha = 0.7$ . Therefore, for a frequency of 800 MHz, the flux density is approximately 1955 Jy, which is 1.52 times as much as the flux density of the example with the phone.

With the help of the equation 2.6.8, the expected antenna temperature can be calculated.  $k$  is the Boltzmann constant and  $A_e$  the effective area of the antenna. The theoretical value of the flux density is  $940 \pm 50$  for a wavelength of 0.21 cm for Tau A [6].

$$\Delta S_v = \frac{2kT_A}{A_e} \quad (2.6.7)$$

$$T_A = \frac{\Delta S_v A_e}{2k} \quad (2.6.8)$$

$$T_{A,theo} = 1.2 \pm 0.07K \quad (2.6.9)$$

Then, a map of Tau A is produced using the telescope to determine an experimental value of  $T_A$ . Different versions of this map can be seen in figure 9.

The mean is normalised to the systemtemperature  $T_{sys}$ , which is also the  $T_A(OFF)$ . With this the noise of the map can be determined:

- $T_{sys} = 11011 \pm 35 \hat{=} 89.8 \pm 0.29 K$

The antenna temperature  $T_A$  on the source position (in the middle of the map) can be determined by scaling it the same way.

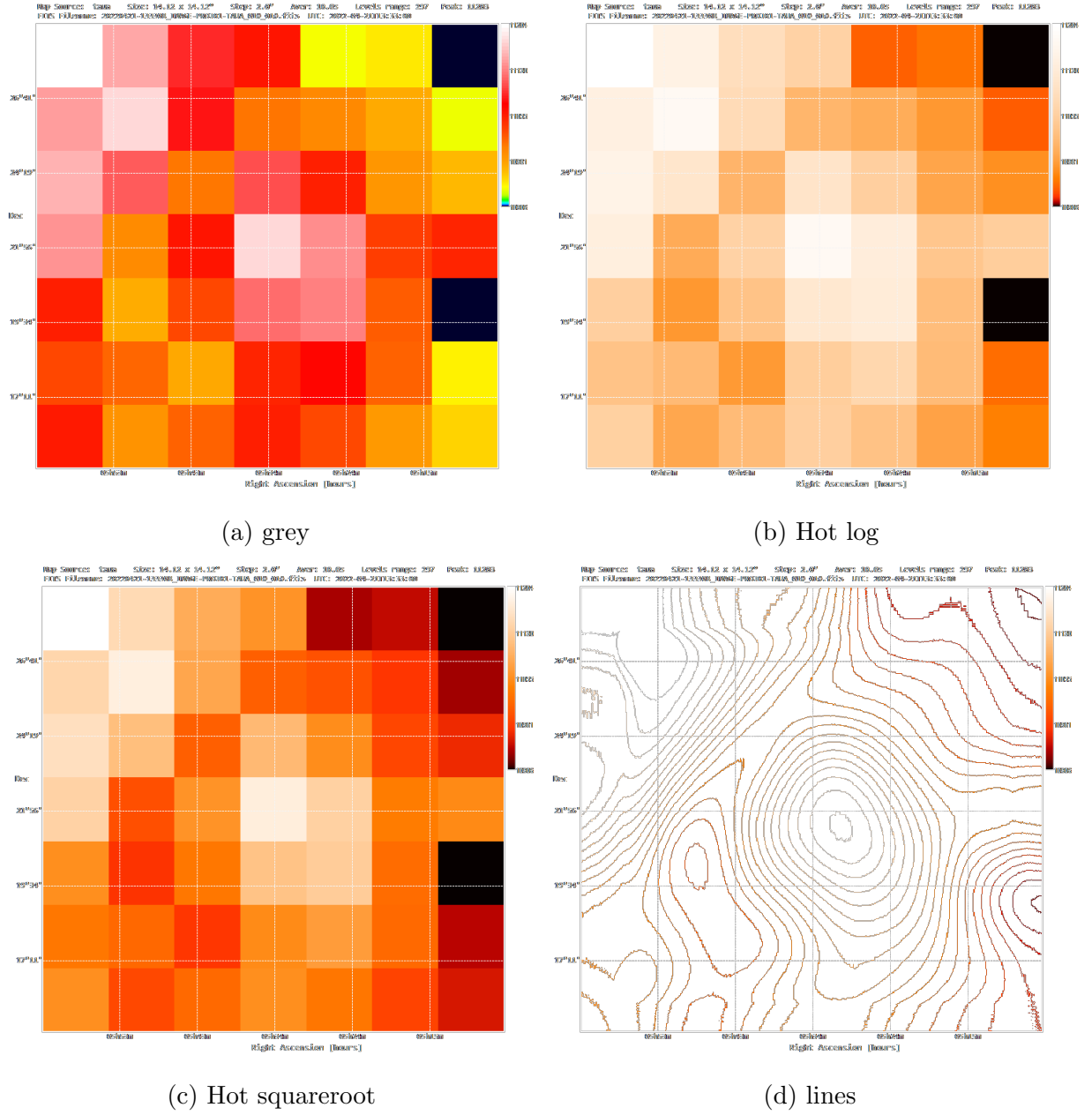


Figure 9: Different version of the map for the taurus region. A maximum is visible in the middle. However, part of the Milky Way is also portrayed on the left top corner.

The antenna temperature is then calculated to be  $1.3 \pm 0.3$  K using equation 2.6.5, which is very close to the theoretical value. Due to the change between on and off positions, the errors can also include variations in the weather, the tracking and the sensitivity of the receiver.

The background is not constant as there is probably the milky way interfering at the top left corner. This can be compared quantitatively by calculating the  $\frac{S}{N}$  ratio:

$$\frac{T_A(ON)}{T_A(OFF)} = \frac{91.1 \pm 0.01K}{89.8 \pm 0.29K} \approx 1.01 \pm 0.28 \quad (2.6.10)$$

For a higher  $\frac{S}{N}$ -ratio the integration time needs to be higher with a relation of  $\frac{S}{N} \propto \sqrt{t}$ . The difference in the theoretical and the experimental value occurs, because when the gain of the receiver fluctuates only around one percent, the value of the temperature already fluctuates around 1K. The actual measurement times have not been evaluated from the log of the measurements.

## 2.7 Observation of neutral hydrogen in the Milky Way (29.03.2022)

This series of measurements was carried out on 29.03.2022.

### 2.7.1 Weather observation on date: 29.03.2022

The weather was dry and a little bit windy. The sky was slightly cloudy and it was about 11 ° C outside temperature. The visibility was good.

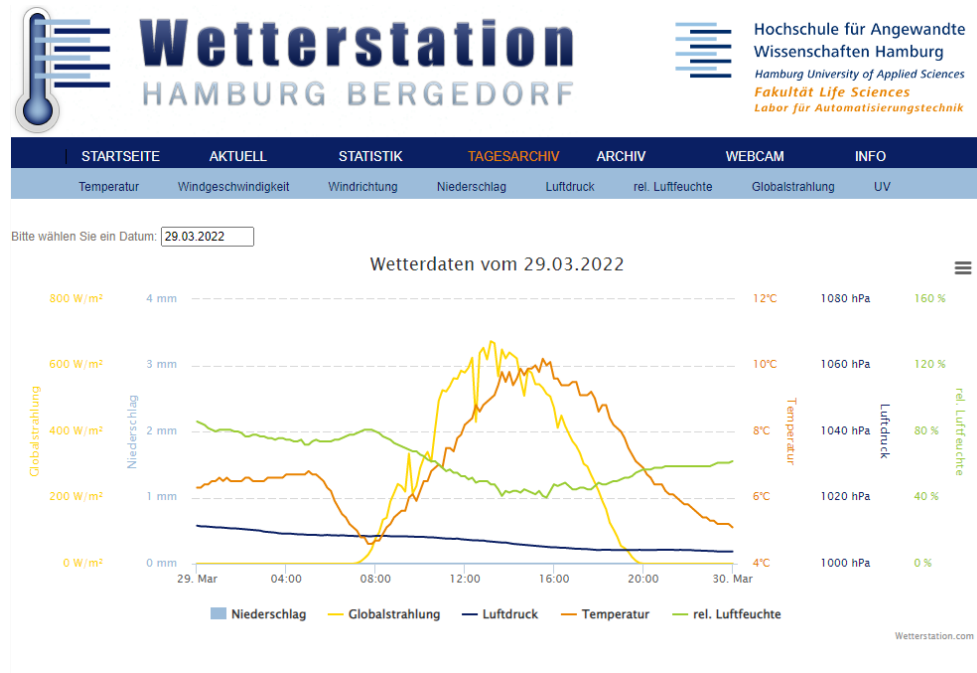


Figure 10: weather observation on 29.03.22

### 2.7.2 Determine frequency setting of the receiver

To determine the radial velocity of the neutral hydrogen, with the centering of the antenna on the 21 cm lines frequency  $\nu_0 = 1420.4$  MHz, the velocity resolution is significant. It is calculated with the following formula:

$$\Delta\nu = 1460.56 \text{ MHz} - 1398 \text{ MHz} = 62.56 \text{ MHz} \quad (2.7.1)$$

$$\delta\nu = \frac{\Delta\nu}{\nu_0} = 0.0611 \text{ MHz} \quad (2.7.2)$$

$$\delta v = c \cdot \frac{\delta\nu}{\nu_0} = 12.8 \frac{\text{km}}{\text{s}} \quad (2.7.3)$$

The SPIDER 300A operates with 1024 channels with a bandwidth of  $\Delta\nu = 62.56$  MHz. This results in a resolution of  $\delta\nu = 0.061$  MHz per channel. One can expect the H1 lines at  $\nu_0 = 1420.4$  MHz in BBC 3 and BBC 11 channels, assuming that the radial velocity is 0 km/s. The frequency range of BBC3 and BBC 11 is each case 7.8125 MHz. The destination is 1420.4 Mhz. Thus, the frequency bands for the right and left pol. must be 367 MHz.

### 2.7.3 Measurement of the 21cm line along the Milky Way plane

To measure the 21 cm lines, it is assumed in a first approximation that the observed H1 gas is on a circular orbit around the Galactic center. Thus the orbital velocity is :

$$v(R)^2 = \frac{GM_R}{R} \quad (2.7.4)$$

Now, along a line of sight with the angle to the Galactic longitude to the Sun-Galactic center connection line ( $R_0$ ), the superimposed emissions of all H1 clouds are observed. Under the condition that the orbital velocities do not increase with increasing distance, the highest velocity is reached at the point where a connecting line from the galactic center to the line of sight has a right angle.

In order to observe the H1 lines and the continuum channels 1 - 6 and 9 - 14 were used to reduce the noise and to use the optimal frequency bands.

To determine the velocity of a spectral line with respect to a reference location, several corrections are required. The orbit of the earth around the sun and the motion of the sun itself have to be subtracted from the observed radial velocity  $V_{\text{obs}}$ . This correction velocity is called  $\Delta V_{\text{LSR}}$  (LSR = local-standard-of-rest).

$$V_{\text{rad}/\text{max}} = V_{\text{obs}} - V_{\text{LSR}} \quad (2.7.5)$$

Then the maximum value of  $V_{\text{LSR}}$  is determined, which is called  $V_{\text{max}}$ , which at the same time represents  $V_{\text{rad}}$ . The value of  $V_{\text{max}}$  can be read in the table 3.

From  $V_{\text{max}}$  we can now calculate the orbital velocity  $V_{\text{rot}}$  (See table 3) , which the hydrogen atoms have when they are at a tuned distance from the galactic center.

$$V_{\text{rot}} = V_{\text{max}} + \omega_0 \cdot R \quad (2.7.6)$$

Whereby the distance of the tangential point from the galactic center is given by the following formula:

$$R = R_0 \cdot \sin(l) \quad (2.7.7)$$

and

$$1kpc = 3.086 \cdot 10^{19} \text{ km}$$

The galactic latitude is set to a constant value of approx. is constantly set to approximately 0, so that the images are taken in the plane of the Milky Way. The On-measurement is done at a galactic Latitude of 15. These measurements are made at different galactic longitude settings between 10 and 90 in steps of 9.

As shown in the figure 11, the value of  $V_{\text{rot}}$  could then be calculated with the  $V_{\text{max}}$  value taken. More plots are in the appendix in fig 19, 20, 21 and 22. The data obtained were then entered in table 3. The orbital velocity  $V_{\text{rot}}$  was then plotted over the distance R in fig 12.

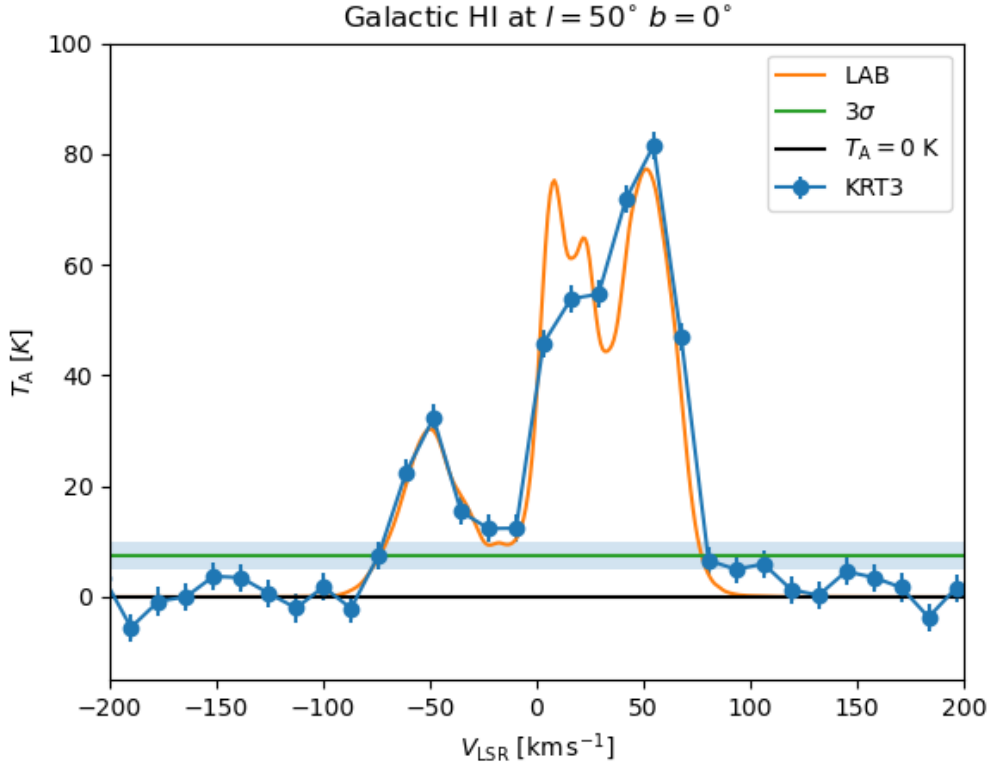


Figure 11: Example spectrum of the recorded 21 cm line as a function of the Radial velocity at a galactic longitude of  $50^\circ$  plotted in blue. The data points result from the averaging of 10 values. The comparison spectrum of the Leiden-Argentine-Bonn (LAB) survey, which is located at the same position, is plotted in orange.

| Gal. Long. (l) | R [m]                | $R_{\text{err}}$ [m] | $V_{\text{max}}$ [m/s] | $V_{\text{max,err}}$ [m/s] | $V_{\text{rot}}$ [m/s] | $V_{\text{rot,err}}$ [m/s] |
|----------------|----------------------|----------------------|------------------------|----------------------------|------------------------|----------------------------|
| 15             | $6.79 \cdot 10^{19}$ | $8.89 \cdot 10^{18}$ | 99670.0                | 7585.0                     | 156617.13              | 15039.90                   |
| 20             | $8.98 \cdot 10^{19}$ | $8.49 \cdot 10^{18}$ | 127530.0               | 4495.0                     | 202855.52              | 11613.39                   |
| 30             | $1.31 \cdot 10^{20}$ | $7.93 \cdot 10^{18}$ | 118260.0               | 9890.0                     | 228271.49              | 16542.46                   |
| 40             | $1.68 \cdot 10^{20}$ | $7.02 \cdot 10^{18}$ | 95260.0                | 8165.0                     | 236592.41              | 14053.85                   |
| 50             | $2.01 \cdot 10^{20}$ | $5.83 \cdot 10^{18}$ | 79990.0                | 13840.0                    | 248501.72              | 18732.28                   |
| 60             | $2.27 \cdot 10^{20}$ | $4.58 \cdot 10^{18}$ | 59390.0                | 3720.0                     | 249904.02              | 7563.93                    |
| 70             | $2.47 \cdot 10^{20}$ | $6.29 \cdot 10^{18}$ | 41270.0                | 12305.0                    | 248091.61              | 17582.96                   |
| 80             | $2.58 \cdot 10^{20}$ | $3.30 \cdot 10^{18}$ | 41300.0                | 16790.0                    | 257957.93              | 18959.70                   |
| 90             | $2.62 \cdot 10^{20}$ | $6.02 \cdot 10^{18}$ | 39220.0                | 1270.0                     | 259242.99              | 1270.00                    |

Table 3: Table for the data values of the rotational curve in figure 12.

Now the rotational velocity  $V_{\text{rot}}$  can be represented as a function of the distance R. In the figure 12, it can be seen that the rotation speed increases sharply up to the radius at about 1.25 kpc. After that, the curve flattens out considerably. In comparison, the barionic velocity continues to decrease with the larger radius. This results in a difference between the barionic and the total velocity. Contrary to the expectation for Kepler systems, the curve is almost constant at larger radii  $R > 2$ . From this it can be concluded that there must be additional mass in the Milky Way, which however cannot be observed.



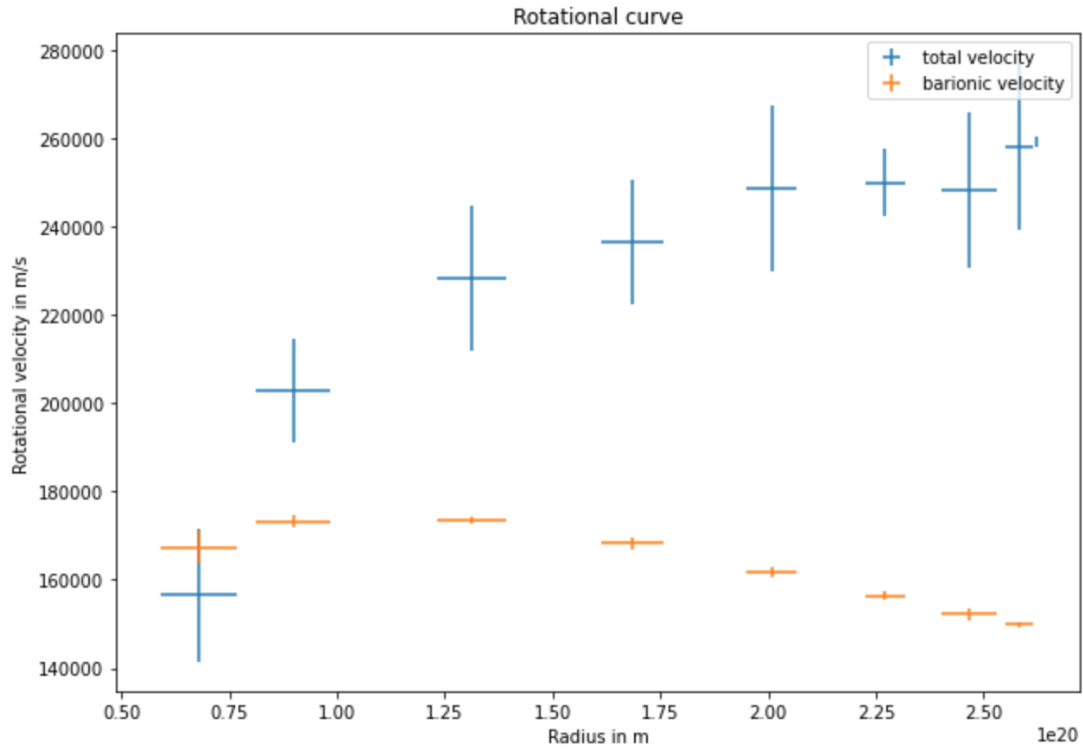


Figure 12: Calculated rotation curve of the total and barionic velocity plotted over the radius from the distance to the Galactic center. The exact values and errors can be found in the table 3.

The curve of the figure 12 can be compared with a curve from literature 13 and it can be seen that there are some similarities of the course of the measured curve with that of the curve from literature. However, the course of the measured curve is shifted to the left by about 2 kpc.

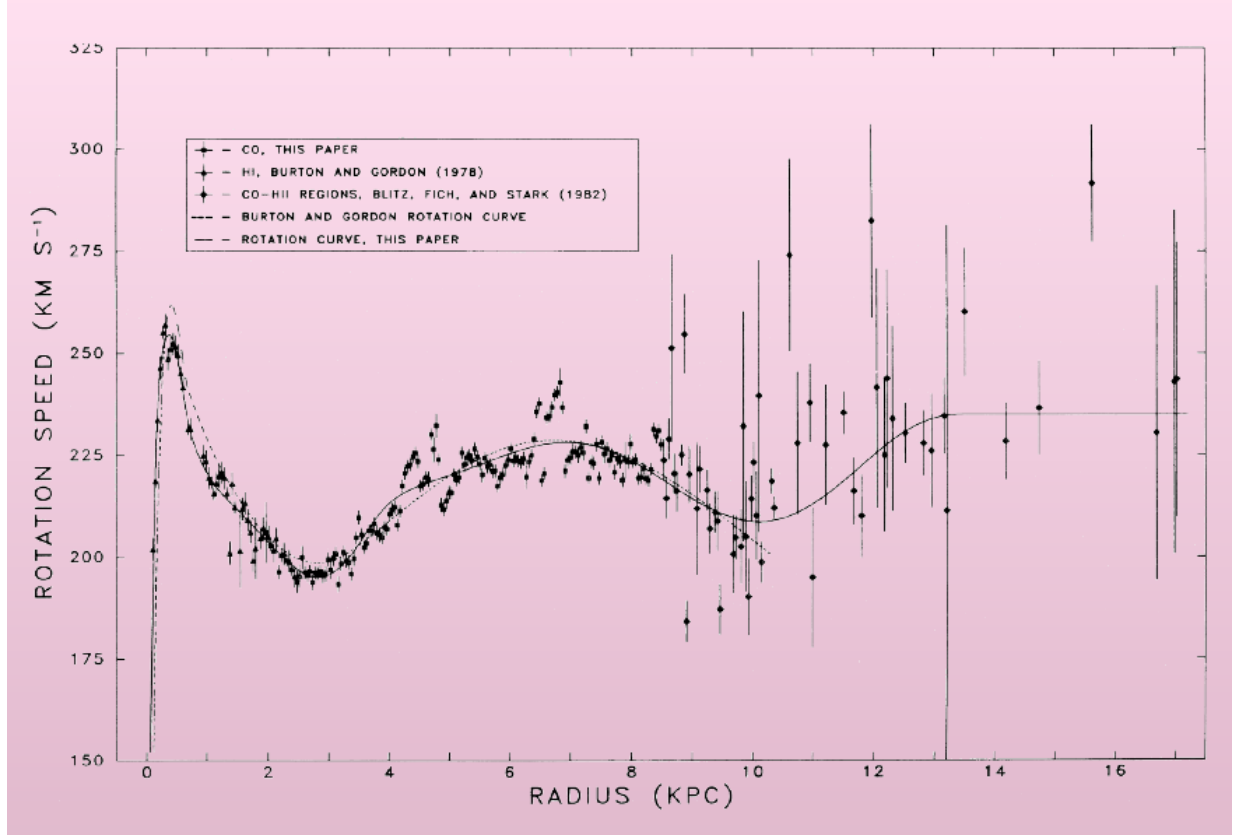


Figure 13: Rotation curve of the Milky Way from literature. Figure taken from [7].

#### 2.7.4 Mass of the Galaxy and distribution of the dark matter

With the calculated orbital velocity and the distance to the center of the Milky Way, the mass of the Galaxy can now be calculated.

$$M_{\text{total}} = 0.234 \cdot 10^{10} \left( \frac{V_{\text{rot}}}{100 \text{ km s}^{-1}} \right)^2 \left( \frac{R}{\text{kpc}} \right) M_{\odot} \quad (2.7.8)$$

Now the mass of the baryonic matter can be subtracted from the total mass and for each distance  $R$  the mass of the dark matter can be calculated. This mass can be plotted as a function over the distance in figure 14 to obtain the mass distribution of the barionic and dark matter. The values and errors can be seen in table 4.

Since the surface mass density of the galactic disk can be described approximately as an exponential function:

$$\sum_d = 1905 \exp^{\frac{-R}{2 \text{ kpc}}} M_{\odot} \text{pc}^{-1} \quad (2.7.9)$$

The rotation curve of the baryonic matter can be calculated with the following formula and is shown in figure 12.

$$v_d = 452 \sqrt{\frac{kpc}{R} [1 - (\frac{R}{2kpc} + 1) \exp^{\frac{-R}{2kpc}}]} km s^{-1} \quad (2.7.10)$$

From the difference of the mass of the total matter and the mass of the barionic matter, now the mass of the dark matter can be determined (see table 4) and plotted as curve 14.

With the formula:

$$\langle \rho_{DM} \rangle = \frac{3}{4} \frac{M_{DM}}{R_0^3 \pi} \quad (2.7.11)$$

And with  $1 GeV cm^{-3} = 1.6 \cdot 10^{-4} J m^{-3}$ , the average dark matter energy density within the solar orbit can now be calculated  $\rho_{DM}(R_0)c^2 = 1.32 \pm 0.12 \text{ GeV cm}^{-3}$ .

| R [m]                | $V_{rot}$ [m/s] | $M_{total}$          | $M_{total,error}$  | $M_{bar}$            | $M_{bar,error}$   | $M_{DM}$              | $M_{DM,error}$       |
|----------------------|-----------------|----------------------|--------------------|----------------------|-------------------|-----------------------|----------------------|
| $6.79 \cdot 10^{19}$ | 156617.13       | $1.26 \cdot 10^{10}$ | $1.65 \cdot 10^9$  | $1.44 \cdot 10^{10}$ | $1.88 \cdot 10^9$ | $-1.18 \cdot 10^{09}$ | $3.54 \cdot 10^{09}$ |
| $8.98 \cdot 10^{19}$ | 202855.52       | $2.80 \cdot 10^{10}$ | $2.65 \cdot 10^9$  | $2.04 \cdot 10^{10}$ | $1.93 \cdot 10^9$ | $0.76 \cdot 10^{10}$  | $4.58 \cdot 10^{09}$ |
| $1.31 \cdot 10^{20}$ | 228271.49       | $5.18 \cdot 10^{10}$ | $3.13 \cdot 10^9$  | $3.00 \cdot 10^{10}$ | $1.81 \cdot 10^9$ | $2.18 \cdot 10^{10}$  | $4.95 \cdot 10^{09}$ |
| $1.68 \cdot 10^{20}$ | 236592.41       | $7.15 \cdot 10^{10}$ | $3.00 \cdot 10^9$  | $3.62 \cdot 10^{10}$ | $1.51 \cdot 10^9$ | $3.53 \cdot 10^{10}$  | $4.49 \cdot 10^{09}$ |
| $2.01 \cdot 10^{20}$ | 248501.7        | $9.41 \cdot 10^{10}$ | $2.173 \cdot 10^9$ | $4.00 \cdot 10^{10}$ | $1.16 \cdot 10^9$ | $5.41 \cdot 10^{10}$  | $3.89 \cdot 10^{09}$ |
| $2.27 \cdot 10^{20}$ | 249904.02       | $1.08 \cdot 10^{11}$ | $2.17 \cdot 10^9$  | $4.21 \cdot 10^{10}$ | $8.45 \cdot 10^8$ | $6.54 \cdot 10^{10}$  | $3.02 \cdot 10^{09}$ |
| $2.47 \cdot 10^{20}$ | 248091.61       | $1.15 \cdot 10^{11}$ | $2.93 \cdot 10^9$  | $4.35 \cdot 10^{10}$ | $1.12 \cdot 10^9$ | $7.17 \cdot 10^{10}$  | $4.04 \cdot 10^{09}$ |
| $2.58 \cdot 10^{20}$ | 257957.93       | $1.30 \cdot 10^{11}$ | $1.67 \cdot 10^9$  | $4.40 \cdot 10^{10}$ | $5.62 \cdot 10^8$ | $8.64 \cdot 10^{10}$  | $2.23 \cdot 10^{09}$ |
| $2.62 \cdot 10^{20}$ | 259242.99       | $1.10 \cdot 10^{11}$ | $1.10 \cdot 10^9$  | $4.42 \cdot 10^{10}$ | $7.52 \cdot 10^8$ | $8.95 \cdot 10^{10}$  | $1.10 \cdot 10^{10}$ |

Table 4: Table for the data values of figure 14 for the mass distribution for different radii.

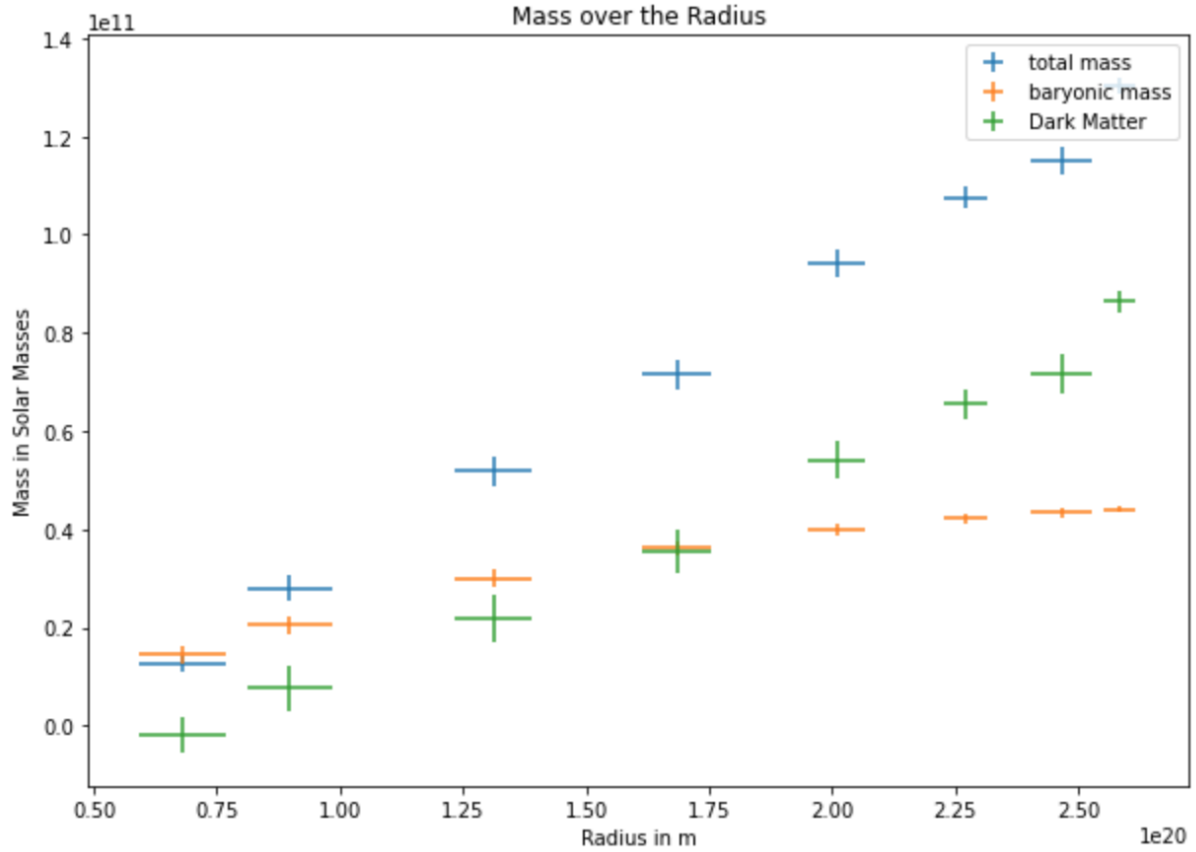


Figure 14: Mass distribution as a function of the radius. The mass distributions of the total mass, barionic mass and dark matter mass are plotted.

### 3 Summary

This section summarises and discusses the calibration and measurement of the H1 21cm lines. First, the calibration, background radiation and spatial resolving power is discussed and then, the measurement and calculation of the mass is summarised.

#### 3.1 Discussion and summary of calibration, background radiation and spatial resolving power

The calibration and characterization were supposed to be done on the first day. However, this had to be done on different days, which also changed the weather conditions. For this, first a reference point in the sky had to be found, where the further measurements could be done. Due to visibility and frequency spectrum the sun was chosen, whose position was determined with a search for the maximum power at the location of the sun.

Overall errors can be attributed to the technical setup of the antenna. The amplification and conversion of the signal can lead to errors. Also, the wind was an influence as well as other weather influences.

Subsequently, the measured total power was recorded the reference value of the sun. Thus, the power TP could be converted to antenna temperature, yielding a factor of  $T_{\text{sys, left}} = 95.5 \pm 13.9$  K for the left-circular-polarized components and  $T_{\text{sys, right}} = 84.0 \pm 16.8$  K for the right-circular-polarized components and  $T_{\text{sun}} = 1551.7 \pm 229.7$  K for the Sun. The relatively high error at right polarization of about 15 percent, at left polarization of about 20 percent, and at 15 percent for the sun, are due to sources of interference and fluctuations at the edge of the measurement range. The influence of the interference sources was counteracted by clever selection of the BBC, so that the H1 measurement was exactly in the frequency corresponding to 21 cm lines. Nevertheless, the error during calibration remained large. The flux density determined was  $S_{\text{vsys, left}} = 74558.3 \pm 8339.9$  Jy,  $S_{\text{vsys, right}} = 65620.6 \pm 13144.4$  Jy and at  $S_{\text{vsun}} = 1211775.3 \pm 179369.8$  Jy.

Then, the background radiation was determined by doing a cross-scan of Orion. For the latitude, it was estimated to be around  $T = 279.59 \pm x$  K, compared to the black body radiation of the earth of  $T_{\text{earth}} = 252$  K. This is closer than expected of a two point interpolation. For the longitude, it is estimated to be around  $T = 88.43 \pm 3.46$  K. However, again this estimation is only using a few of the measured data points and is therefore very subjective.

After that, the spatial resolving power is determined by doing a cross-scan across the sun. The theoretically expected FWHM is  $4.0909^\circ$ . The average of all four measurements (latitude, longitude and both circularisations) estimated a FWHM of  $(5.34 \pm 0.08)^\circ$ . The difference between the theoretical and the experimental value could be due to the error in the calculation of the temperature and the difference in the temperature in the sun between the two measurements.

Furthermore, the sensitivity was determined. The theoretical value of the lowest variable flux density is calculated to be  $\Delta S_v = 283.93 \text{ Jy}$ . The experimental value of the antenna temperature when directed at Tau A is  $1.3 \pm 0.3 \text{ K}$ , compared to the theoretical value of  $T_{A,\text{theo}} = 1.2 \pm 0.07 \text{ K}$ . A map of Tau A was drawn and portrayed in four different variations in order to calculate those. Part of the Milky Way is visible in the corner. The signal to noise ratio was also calculated with a difference in the theoretical and experimental value due to different integration times and the fluctuations in the gain of the receiver.

### 3.2 Discussion of the H1 21cm line

During the main measurement in the experiment, the 21 cm lines of atomic hydrogen were measured. Its spectra were measured at galactic lengths from  $15^\circ$  to  $90^\circ$  galactic longitude to the galactic nucleus. With the help of the doppler effect and the accompanying frequency shift, which resulted from the relative velocity of the H1 atoms to the telescope, the radial velocity of the hydrogen at different distances could be calculated. At the tangential point, from the line of sight orthogonal to the galactic center, the velocity becomes Maximal  $V_{max}$  and it is possible to calculate with it the total rotational velocity  $V_{rot}$ , which was then plotted as a function over the distance and compare with the velocity of the baryonic matter. (see figure 12). It can be seen that the shape of the curve in the range  $6.79 \cdot 10^{19}$  (2.2 kpc) to  $2.62 \cdot 10^{20}$  (8.5 kpc) approximates that from the literature. Furthermore, it can be seen that the rotation curve is constant from nearly  $2.01 \cdot 10^{20} \text{ m}$ .

This deviation from the literature can be explained by the strong sources of interference that are reflected in the calculated error (in table 3 ). The relative errors were large up to 10 percent, but almost constant over the whole distance.

First, there was a strong radio signal that was not transmitted by the target source but must have originated in the relative vicinity of the telescope. To avoid this interference, some channels were excluded from the measurement, as already explained in the section 2.7.2. Other sources of interference could have been the weather. Since there were light gusts of wind on this day, the telescope may have shaken a bit and distorted the measurement.

Furthermore, there is a region in the Milky Way called the Bulge Region, which lies at  $R < 3 \text{ kpc}$ , where the measurements were mainly made. This region is bar-shaped, in which the hydrogen atoms do not move in a circular path as assumed. Thus the assumed formula 2.7.6 is only an approximation to the actual motion of the hydrogen.

From the rotational velocity then, with the formula 2.7.8, the mass of the Milky Way could be calculated over the distance R (table 4). Where at the largest distance  $R = 2.62 \cdot 10^{20} \text{ a}$

total mass of  $1.10 \cdot 10^{11}$  and a baryonic mass of  $4.42 \cdot 10^{10}$  was calculated. The calculated results of total mass, baryonic mass, and dark matter mass plotted as a function of distance 14. At the distance  $6.79 \cdot 10^{19}$  the mass of baryonic matter is above the total mass, which is an error. The relative errors are about 10 percent again and can be explained by the errors in the calculation of the rotational velocity  $V_{rot}$ , since they have propagated.

It can be seen that the total mass of the Milky Way is increasing linearly, although the rotation speed remains almost constant. This suggests that there must be other matter in the Milky Way that does not interact with electromagnetic radiation and thus cannot be measured by our telescope. This matter is called dark matter because of its properties. It is assumed that it is located in a spherical halo around the Milky Way. The calculated curve of dark matter mass is almost parallel to the curve of total mass, where mass of baryonic matter decreases with distance. This means that the density distribution of the dark matter increases over the distance and that of the baryonic matter decreases. This agrees with the theoretical idea that the dark matter is in a halo around the galaxy and with the observation of the loud matter, which concentrates in the core and decreases with larger distance in the spiral arms.

Now the determined total mass at  $R = 2.62 \cdot 10^{20}$  m (8.5 kpc) can be calculated to  $3.08 \cdot 10^{21}$  m (100 kpc) and compared to the literature value of  $10^{12}$  [8]. This yields a value of  $\approx 1.57$ . Now by the difference of the total mass and the baryonic mass the mass of the dark matter was calculated and its energy density was calculated. The determined value of  $\rho_{DM}(R_0)c^2 = 1.32 \pm 0.12 \text{ GeV cm}^{-3}$ . can be compared with the value from literature  $\rho_{DM}(R_0)c^2 = 0.43 \pm 0.11 \text{ GeV cm}^{-3}$  [8]. This results in a deviation of  $0.89 \text{ GeV cm}^{-3}$ .

## References

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- [3] Hochschule für Angewandte Wissenschaften Hamburg. Wetterstation Bergedorf- Tagesarchiv. 04 2022.
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- [7] Dieter Engels Hamburger Sternwarte. HS1 Radioastronomie - Hintergrund.
- [8] Dieter Engels Volker Heesen Hamburger Sternwarte. HS1 Radioastronomie Versuchsmappe.



## A Appendix

|                                                                                                                     |          |
|---------------------------------------------------------------------------------------------------------------------|----------|
| Measurement Log                                                                                                     | 21.04.22 |
| weather: cloudy, see weather data                                                                                   |          |
| 5.1.1. Determination $T_{\text{sys}}$                                                                               |          |
| 5.2 Bestimmung räumliches Auflösungsvermögen (Sonne)                                                                |          |
| ↳ 4 Plots: - Latitude right (FWHM)<br>- Latitude left (FWHM)<br>- Longitude right (FWHM)<br>- Longitude left (FWHM) |          |
| 5.1.2. Hintergrundstrahlung (Orion A)                                                                               |          |
| ↳ 4 Plots: - Latitude right<br>- Latitude left<br>- Longitude right<br>- Longitude left                             |          |
| ↳ Milkyway can interfere                                                                                            |          |
| 5.3.2. Bestimmung Empfindlichkeit                                                                                   |          |
| ↳ Map of Taurus region<br>- grey<br>- hot log<br>- hot sqrt                                                         |          |
| ↳ Milkyway                                                                                                          |          |

Figure 15: Measurement Log 21.04.2022

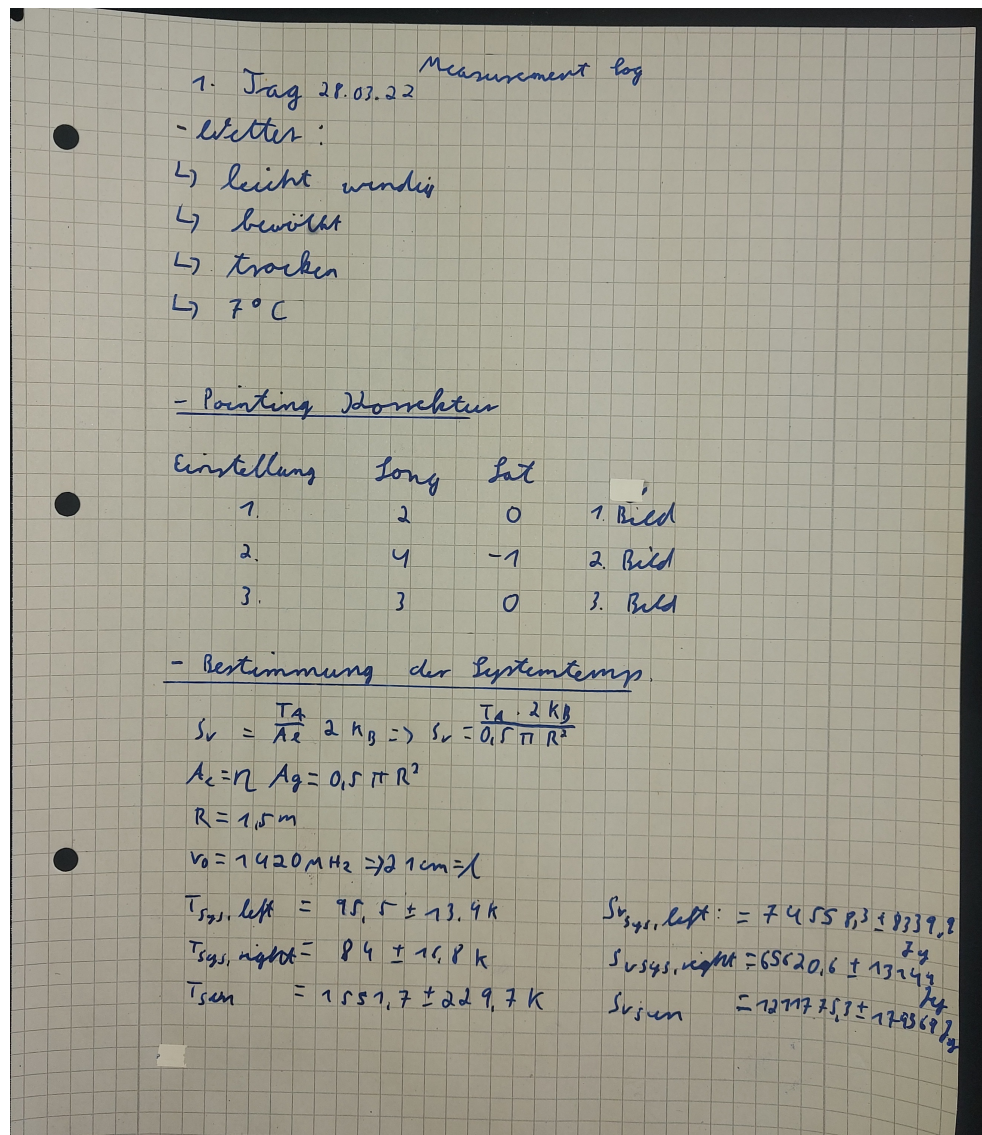


Figure 16: Measurement Log 28.03.2022

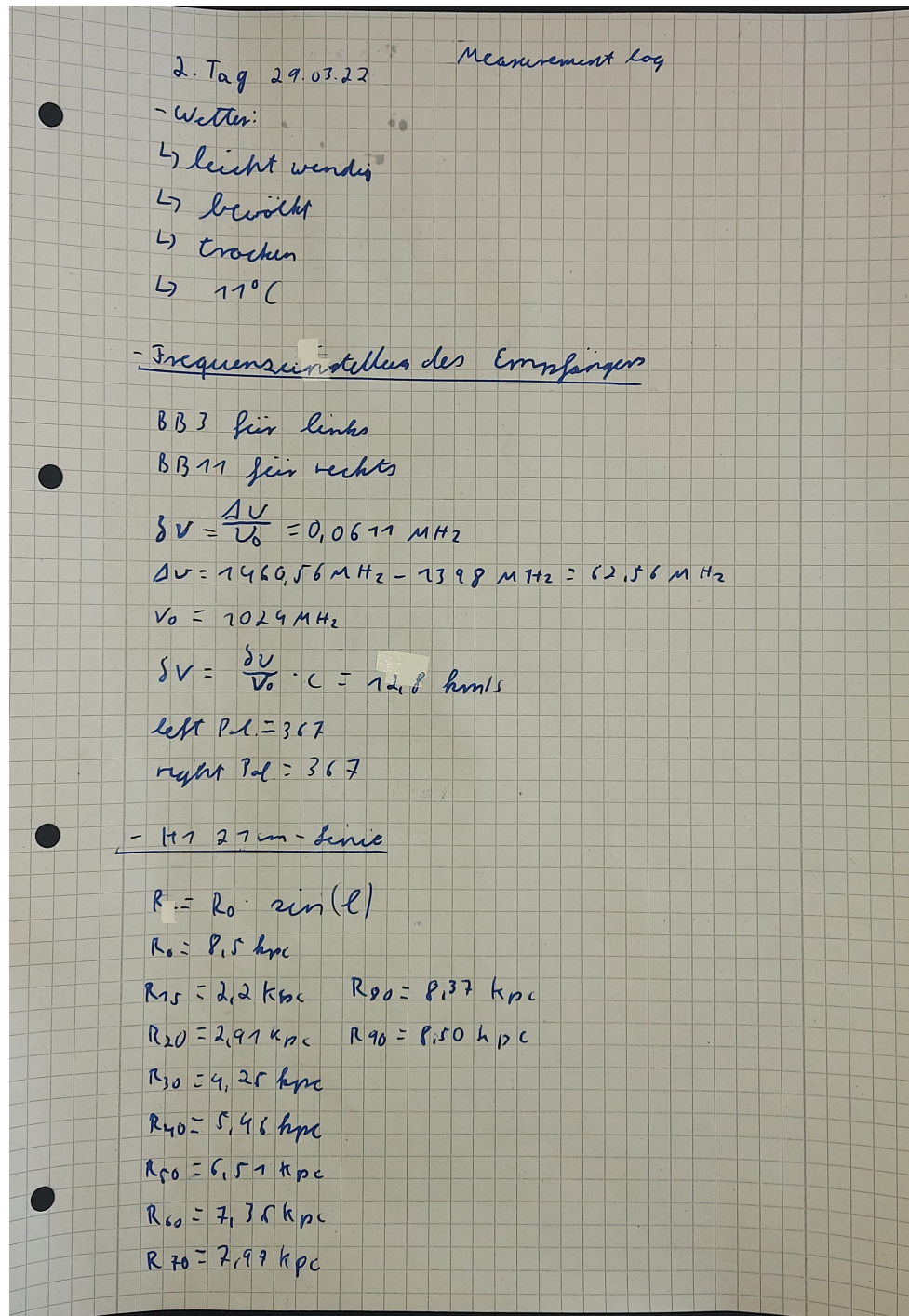
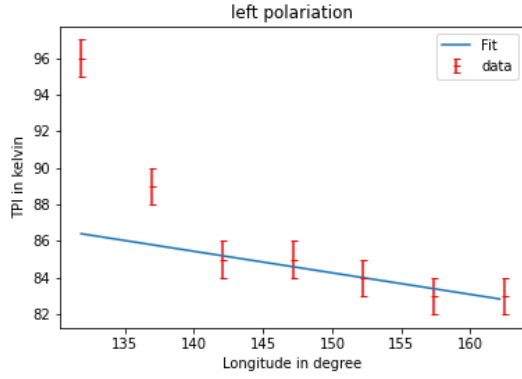
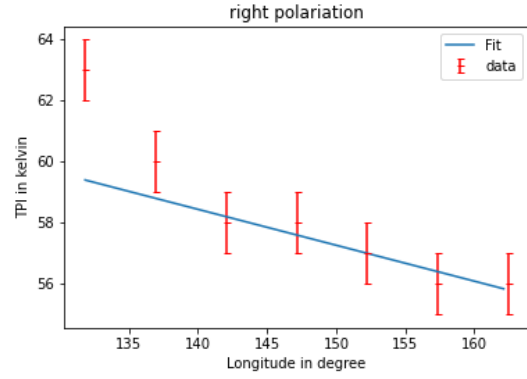


Figure 17: Measurement Log 29.03.2022

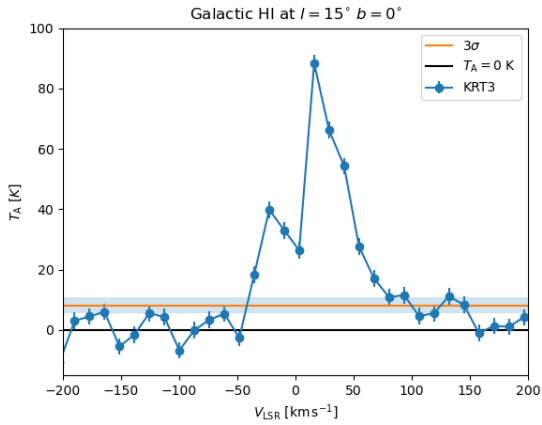


(a) Other scaling: Orion Longitude left circular polarized

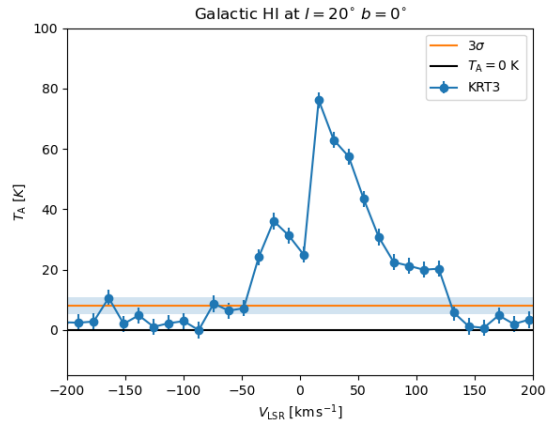


(b) Other scaling: Orion Longitude right circular polarized

Figure 18: Other scaling: Orion

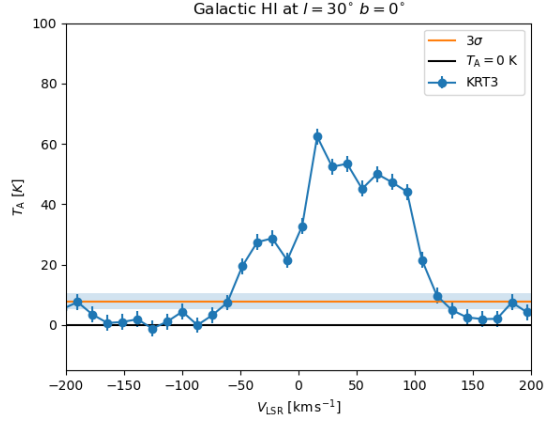


(a) velocity spectra longitude 15

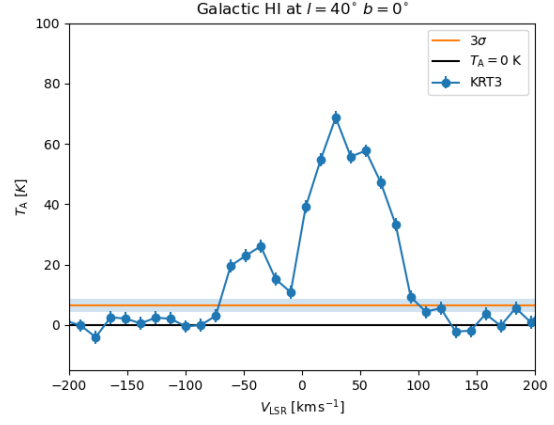


(b) velocity spectra longitude 20

Figure 19: velocity spectra

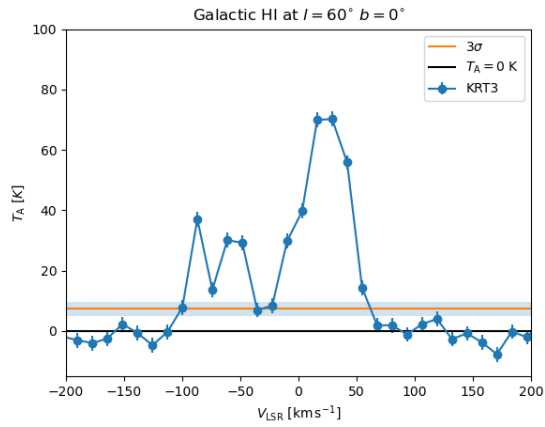


(a) velocity spectra longitude 30

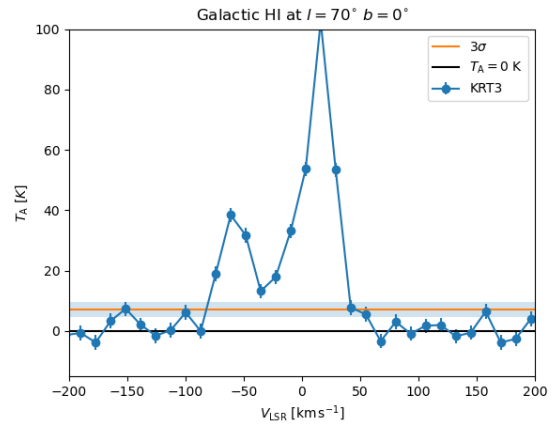


(b) velocity spectra longitude 40

Figure 20: velocity spectra

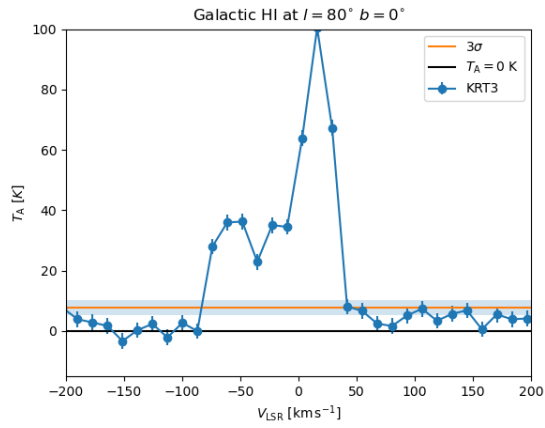


(a) velocity spectra longitude 60

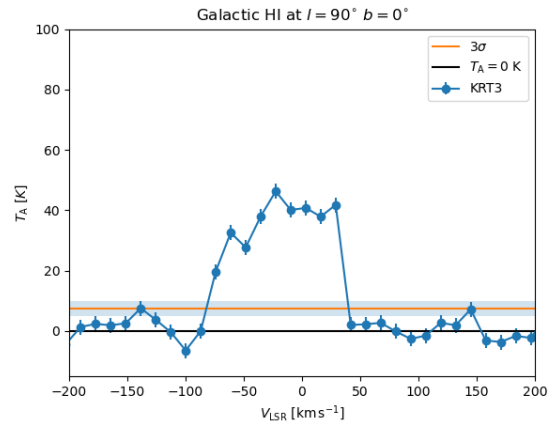


(b) velocity spectra longitude 70

Figure 21: velocity spectra



(a) velocity spectra longitude 80



(b) velocity spectra longitude 90

Figure 22: velocity spectra